

# STRATEGIC PRODUCT PLACEMENT ANALYSIS

## Problem Statement

### Introduction

Retail businesses handle large volumes of sales and customer-related data, but converting this data into meaningful product placement decisions is often difficult. The position of a product inside a store can influence product visibility, customer attention, and overall sales performance. Retailers therefore need a clear analytical approach to understand how product category, product position, customer demographics, foot traffic, pricing, seasonal trends, and promotions affect sales volume.

### Problem Statement

Retailers often lack an interactive and consolidated visualization system for evaluating the effectiveness of product placement. Traditional reports and raw datasets make it difficult to compare product performance across aisles, end-caps, and front-of-store locations. They also make it challenging to identify relationships between sales volume and factors such as consumer demographics, foot traffic, product price, competitor price, season, and promotional activity.

As a result, retail decision-makers may not easily identify high-performing product categories or strategic placement areas. This can lead to reduced product visibility, inefficient placement decisions, and missed opportunities to improve sales performance.

### Key Challenges

- Difficulty in comparing average sales volume across different product categories.
- Limited visibility into the effect of aisle, end-cap, and front-of-store product positions.
- Difficulty in understanding sales patterns across consumer demographic groups.
- Lack of clear analysis of foot traffic and its relationship with average sales volume.
- Difficulty in comparing product price with competitor price.
- Limited understanding of seasonal and promotional effects on product sales.
- Absence of a single interactive dashboard for exploring retail placement insights.

### Need for the Project

There is a need for an interactive data visualization solution that transforms retail data into clear and actionable insights. The solution should enable users to analyze sales volume, product categories, placement positions, consumer demographics, foot traffic, pricing, promotions, and seasonal trends through visual dashboards and an analytical story.

### Proposed Analytical Direction

The Strategic Product Placement Analysis project uses Tableau to create an interactive dashboard and story for retail data analysis. The visualizations help compare product performance, study customer purchasing patterns, evaluate placement effectiveness, and identify strategic product positioning opportunities. The Tableau views are integrated into a Flask and Bootstrap-based web application to provide a structured and accessible interface for viewing the analysis.

## **Expected Outcome**

The project is expected to help retail decision-makers understand product placement performance and identify patterns that can support better positioning strategies. By presenting complex retail data through interactive visualizations, the solution aims to improve insight generation and support data-driven decisions related to product visibility and sales performance.

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